Neural network hand-in

Multiclass prediction (1)

How do we go from a list of probabilities to an actual prediction?

Discuss: What should be our predicted digit if our “black box” returns:

a.

```
Digit:         0   1   2   3   4   5   6   7   8   9

Probability: .01 .01 .03 .05 .02 .06 .01 .10 .65 .06
```

b.

```
Digit:         0   1   2   3   4   5   6   7   8   9

Probability: .03 .02 .05 .29 .02 .27 .03 .04 .18 .07
```

Answer:

Multiclass prediction (2)

Discuss: What is a good “black box” in this case (for the MNIST data)?

Come up with a description both in terms of the class probabilities $\widehat{p}$-s and the final digit predictions $\hat{y}$.

Answer:

Neural networks (1)

Calculate the value at the first hidden node h_1, using input node values:

$$x_1 = 0.8,\ x_2 = 0.4,\ x_3 = 0.6.$$

Use the following weights and bias:

$$w_1 = 0.5,\ w_2 = -1.0,\ w_3 = 0.5,\ b = -0.4.$$

The value h_1 is defined by the ReLU activation function:

$$z_1 = \sum_{i=1}^{3} w_i z_i + b,\ \ h_1 = \mathrm{ReLU}(z_1) = \max(0, z_1)$$

Answer:

Neural networks (2)

How many unknown parameters (weights w and biases b) have we introduced so far?

Answer:

Neural networks (3)

Calculate the class probability at the 10th output node $\widehat{p}_9$, using hidden node values:

$$h_1 = 0.8,\ h_2 = 0.5.$$

Use the following weights and bias:

$$v_1 = 1.0,\ v_2 = 0.4,\ b = 0.0.$$

Use the following 9 output scores for the digits 0, 1, to 8:

$$z_0 = 0.2,\ z_1 = 0.5,\ z_2 = -0.4,\ z_3 = 0.8,\ z_4 = 0.1,$$

$$z_5 = 0.6,\ z_6 = -0.2,\ z_7 = 0.3,\ z_8 = 0.7.$$

Answer:

Neural networks in Python

We have seen an example of a neural network with one hidden layer with 32 nodes and the ReLU activation function. In canvas you will find python code for this.

Step 1

Download 4 files from canvas: the notebook mnist_neuralnet.ipynb and 3 datasets mnist_train.npz, mnist_validation.npz, mnist_test.npz.

Use the NTNU jupyterhub (jupyter.ntnu.no), or run python locally (might be faster).

Run the code that is given, which uses 20 iterations as a stopping criterion. Why do you get a warning?

What is the accuracy on the* **training set** and what is the accuracy on the **validation set**?

Answer:

Step 2

Decide on **two** other architectures:

- Adding a second hidden layer?
- Changing the number of nodes in the hidden layer(s)?
- A different activation function? (ReLU, sigmoid or tanh)

What are your architectures?

What are their accuracies on the **validation set**?

Answer:

Step 3

Choose one model, which one? How did you choose?

Report the accuracy of your chosen model on the **test set**.

Plot the confusion matrix for the **test set**. Which digit was most difficult for the model to predict correctly? What did the model predict instead?

Answer: